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Roll No.

188543

Level 4, 2nd Sem / DVOC (Medical Imaging Tech., SD)
Subject : Digital Electronics

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Very short answer Questions. Attempt all ten questions. (10x1=10)

Q.1 Define Digital Signal

Q.2 In a Boolean algebra $X+X=$ _____

Q.3 Draw the symbol of NAND gate.

Q.4 Write the truth table of Half Adder

Q.5 How many output data lines are there in a 1:32 Demultiplexer?

Q.6 Define Latch

Q.7 How many Flip Flops are required in MOD 10 Counter?

Q.8 Expand LED

Q.9 Define Encoder

Q.10 Expand ROM

SECTION-B

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Differentiate between Analog and Digital System.

Q.12 State and explain Demorgan's Theorem.

Q.13 Implement OR & AND gate using NOR gates only.

Q.14 Explain in brief about the Truth Table, Boolean Expression and logic diagram of Half Adder.

Q.15 Explain Serial in Parallel Out Shift Register with diagram.

Q.16 Explain 7 segment Display in brief.

Q.17 Differentiate between RAM & ROM.

Q.18 Explain Encoders with diagram.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Define Demultiplexer. Design a 1:8 DEMUX by using truth Table and logic diagram.

Q.20 Explain the working of 4 bit Synchronous Counter with the help of a Truth Table and Pulse diagram.